

SW Engineering CSC 648/848 Fall 2025

Gator Learn

Team #5

Team Lead	Milo Pesce Ares	mpesceares@sfsu.edu
------------------	-----------------	---------------------

GitHub Lead	Hill Kalathiya
Frontend Lead	Enrique Joachim Liganor
Backend Lead	Jonah Hammond
Support	Samantha Chombo-Rodriguez

Milestone 2 Part 1

Date submitted	Date revised
11/1/2025	-

1 Executive Summary

Ever had an experience where you thought "Oh no, I have an exam in a week, and I don't understand the material!" but don't know where to find support? Gator Learn has your back!! Gator Learn is a student-led academic support platform that's designed to empower San Francisco State University students through accessible, affordable and personalized tutoring exclusively for SFSU students, by SFSU students. Our platform fosters a collaborative learning environment built on shared academic experiences and community trust. We offer one-on-one and group tutoring across a wide range of SFSU courses including STEM, Humanities, Business, Writing, and general education requirements! Our mission is to enhance student success, confidence and retention by making academic support convenient and relatable. Sessions are available in-person and online with flexible scheduling to fit the demands of university life. Beyond coursework help, tutors provide study strategies, exam preparation, and mentorship to help students thrive long term!

Gator Learn! Helping Gators support Gators!

2 Data Items and Entities

Key Data Entities

1. **UserAccount:**

Description: Stores basic information for every individual using the platform , including students, tutors, and admins.

Attributes:

1. Name
2. E-mail
3. Password
4. Photo (Optional)

5. Description (Optional)
6. Pronouns (Optional)

Usage: Used for login, access control, and permission management. Acts as the foundational link between all user-related data (profiles, sessions, messages).

2. **TutorListing:**

Description: Represents a tutor's service offering

Attributes:

1. Student Name
2. Subject
3. Course(s)
4. Price per hour
5. Available time(s)
6. Resume (CV)
7. Description
8. Photo (Optional)
9. Tutoring Video Sample (Optional)
10. Live (True/False)

Usage: Displayed to students during tutor search and booking. It helps students to compare available tutors based on price, subject, and timing before making a reservation.

3. **Message:**

Description: A one-way communication channel where a student sends a message to a listing owner

Attribute:

1. Student Name
2. Listing Owner Name
3. Tutor Listing
4. Message Content
5. Date and Time
6. Phone Number (Optional)

Usage: Used by the listed student who is getting the tutoring request to get the information about the student, including phone number, and a message.

4. Subject:

Description: Represents an SFSU academic subject that tutors can teach and students can search for.

Attribute:

1. Subject Name
2. Category
3. Description

Usage: Used to categorize tutor listings and organize search filters by subject area. Supports subject-based matching during tutor listing browsing.

5. Course:

Description: Represents an SFSU academic course that tutors can teach and students can search for

Attribute:

1. Course Name
2. Subject
3. Course Number
4. Description

Usage: Used to categorize tutor listings and organize search filters by course. Supports course-based matching during tutor listing browsing.

6. Feedback:

Description: Captures feedback from students after a tutoring session, including star rating and written comments.

Attribute:

1. Student Name
2. Listing Owner Name
3. Rating
4. Comment
5. Posted Date

Usage: Displayed on tutor profiles to help future students make informed decisions.

3 Functional Requirements

Priority 1

Unregistered Users

1. Unregistered users shall be able to register on a dedicated registration page with required name and @sfsu.edu email and optional photo, description, and pronouns
2. Unregistered users shall be able to look at home page, team page, and browsing page
3. Unregistered users shall be able to search and sort tutor listings by name, time, price, SFSU subject, and SFSU course number

Registered Users

4. Registered users shall inherit all unregistered user's requirements and permissions
5. Registered users shall be able to login on a dedicated login page
6. Registered users shall be able to logout
7. Registered users shall be able to one-way reply to tutor listings with an in-site messaging system.
8. Registered users shall be able to submit tutor listing applications with required title, description, available weekly times, price, and CV and optional photo and video for post approval
9. Registered users shall be able to use a dashboard that contains messages received or sent and their pending or approved tutor listings.
10. Registered users shall be able to delete their tutor listings or applications from dashboard

Administration

11. Administrators shall inherit all registered user's requirements and permissions
12. Administrator accounts shall be assigned manually by website manager
13. Administrators shall be required to review all tutor listings before they go live
14. Administrators shall be able to remove any registered user or tutor listing

Priority 2

Registered Users

15. Registered users shall be able to edit account information after registering
16. Registered users shall be able to edit their tutor listings after approval.
17. Registered users shall be able to delete their messages

Priority 3

Unregistered Users

- 18. Unregistered users shall have their own page
- 19. Unregistered users shall be able to get help from a FAQ/Help page

Registered Users

- 20. Registered users shall be able to reset their password
- 21. Registered users shall be able to report listings and messages
- 22. Registered users shall be able to provide feedback on listings
- 23. Registered users shall be able to visit the about pages of other registered users

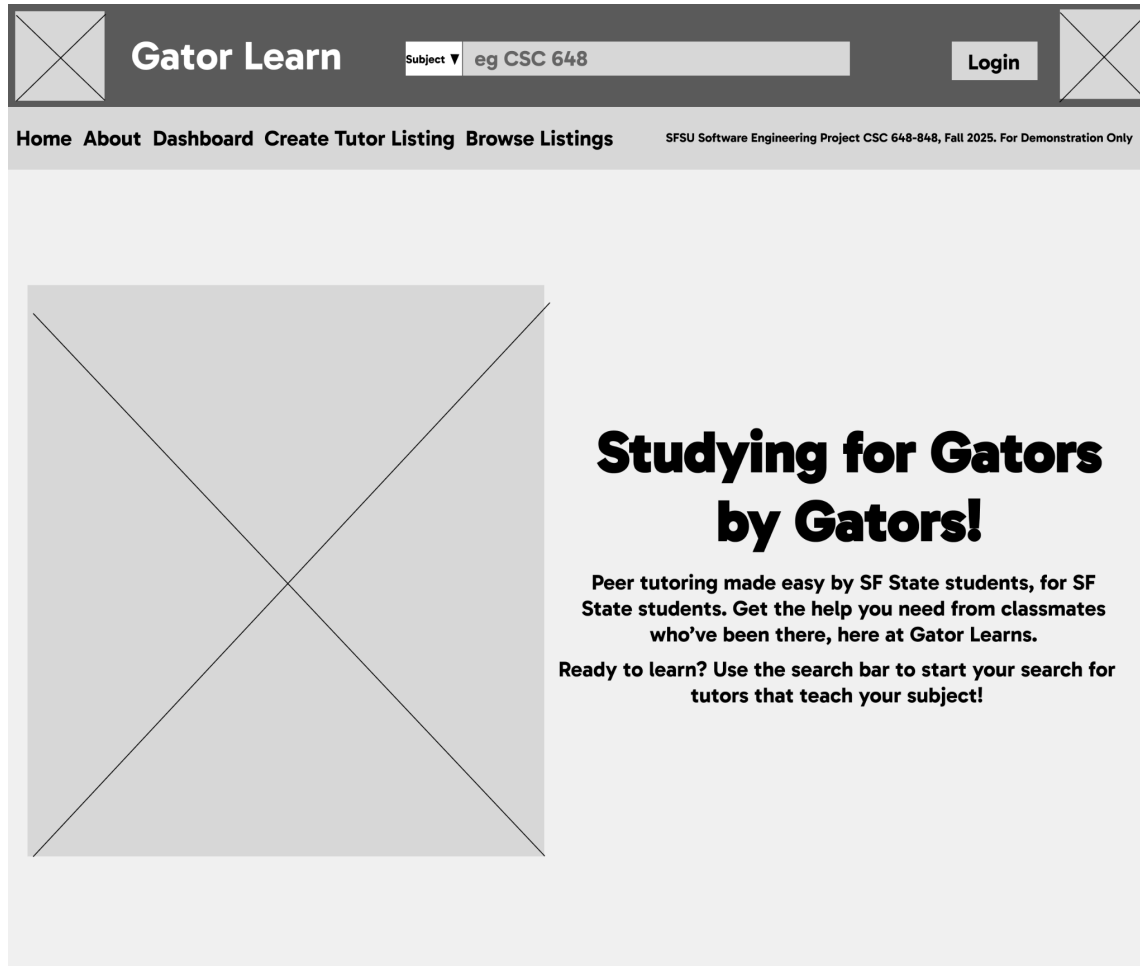
Administration

- 24. Administrators shall be able to view website analytics

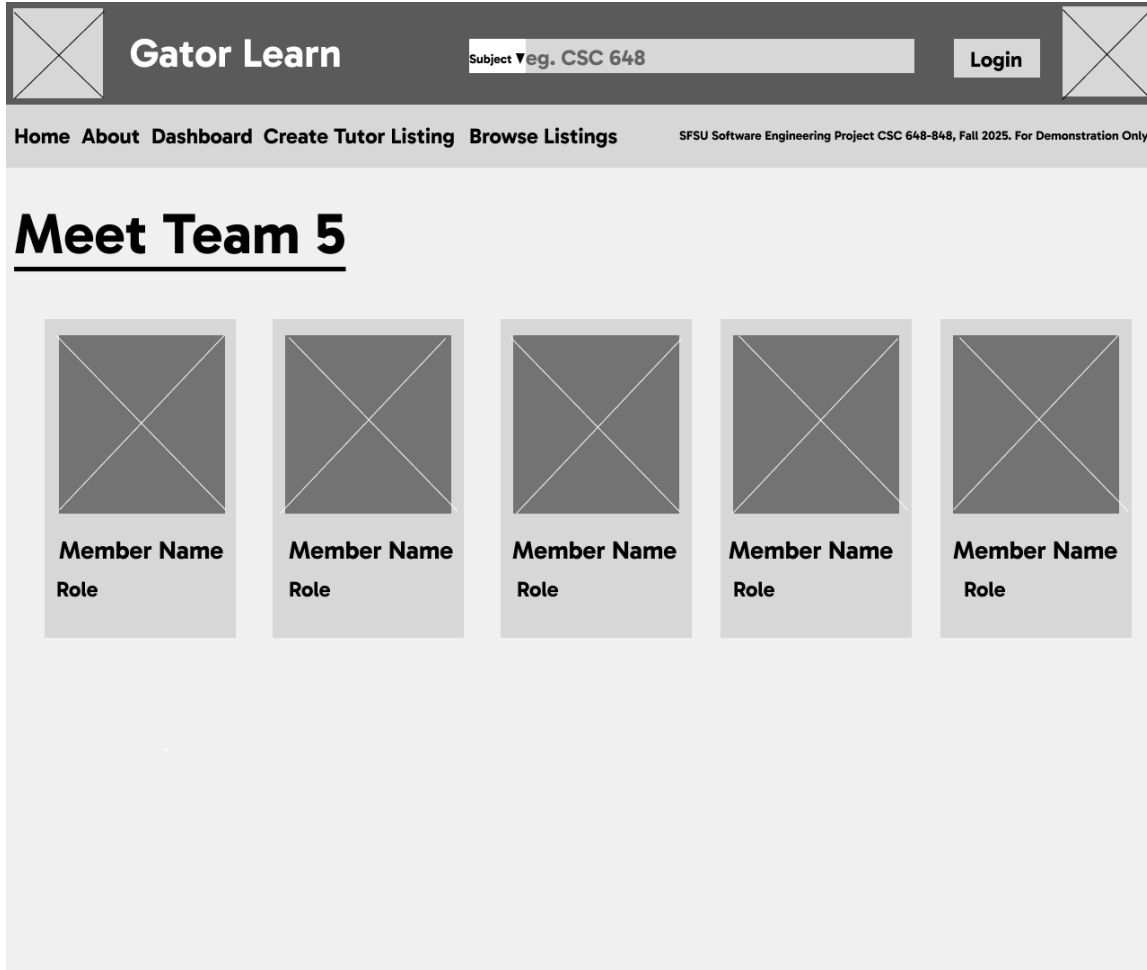
4 UI Storyboards

Use Case 1: Student browses and registers for site

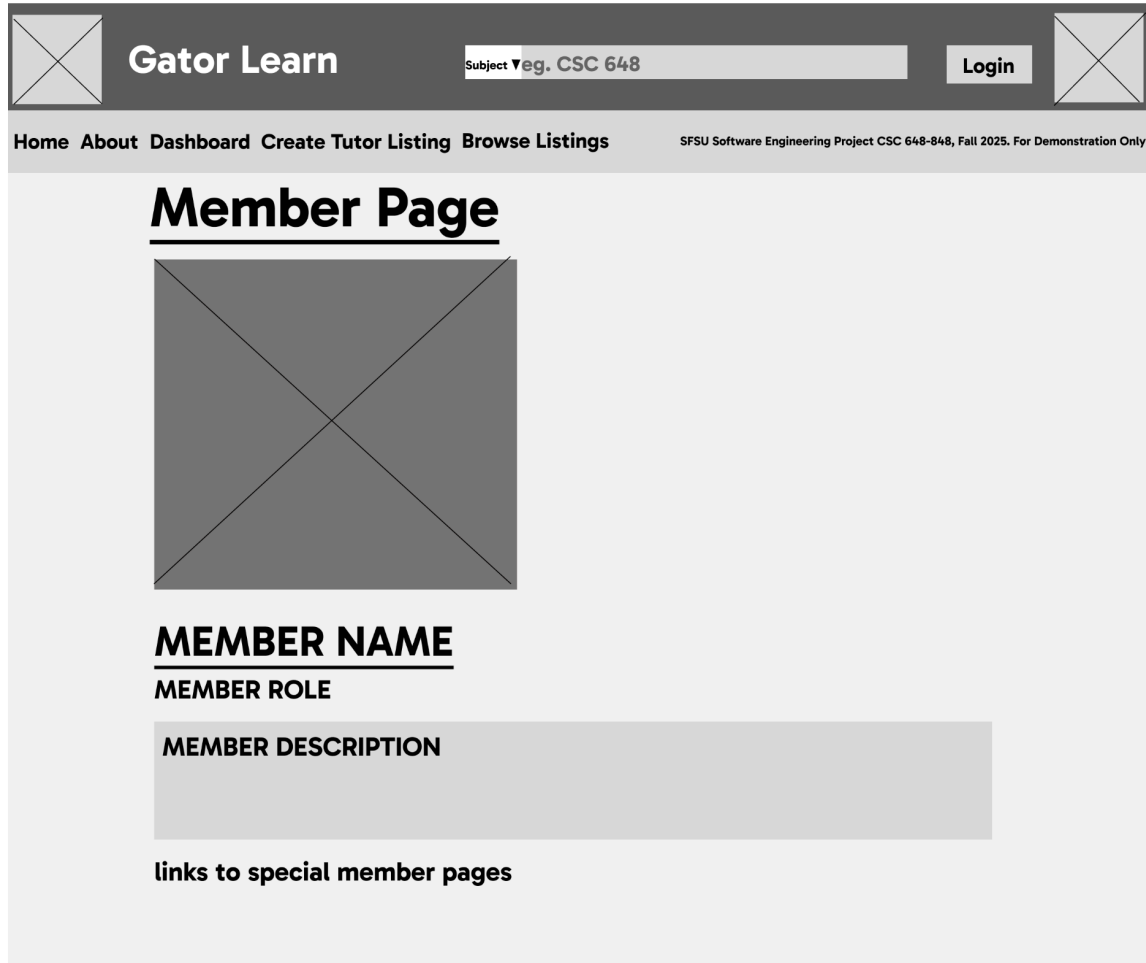
Alex scrolls through the site and browses all the pages available without registering.



Alex sees that the site was made by SFSU students, too, and is able to click on the team page information without registering.



He clicks on one member's image and learns a bit more about them.



He changes pages through the drop down again and browses through the tutor listings.

Gator Learn

Subject ▼ eg. CSC 648

Login

[Home](#) [About](#) [Dashboard](#) [Create Tutor Listing](#) [Browse Listings](#) SFSU Software Engineering Project CSC 648-848, Fall 2025. For Demonstration Only

Current Listings

Found 10 tutors

Sort By Price: \$ → \$\$\$\$ v

Tutor Name

Tutoring for:

Price: \$/hr

Times Available:

[View more details>](#)

MESSAGE TUTOR

Tutor Name

Tutoring for:

Price: \$/hr

Times Available:

[View more details>](#)

MESSAGE TUTOR

Tutor Name

Tutoring for:

Price: \$/hr

Times Available:

[View more details>](#)

MESSAGE TUTOR

Tutor Name

Tutoring for:

Price: \$/hr

Times Available:

[View more details>](#)

MESSAGE TUTOR

Alex tries to send a message without registering. The site sends him to sign up after he has written his message to the tutor.


Gator Learn

Subject ▼

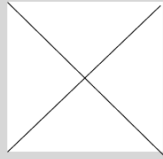
Login

[Home](#)
[About](#)
[Dashboard](#)
[Create Tutor Listing](#)
[Browse Listings](#)

SFSU Software Engineering Project CSC 648-848, Fall 2025. For Demonstration Only

Current Listings

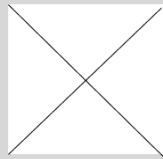
Sort By Price: \$ → \$\$\$\$ ▼



Message : TUTOR NAME for CLASS NAME/NUM

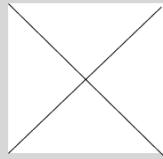
Preferred Contact method:

Discord : Username123 or SFSU EMAIL



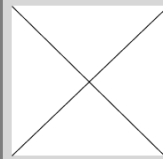
Comments:

I have a midterm next tuesday. Can we meet online?



Cancel

Send Message



Tutor Name

Tutoring for:

Price: \$/hr

Times Available:

[View more details>](#)

MESSAGE TUTOR

Alex then signs up, and credentials are confirmed via SFSU email. His message gets sent with his sign up info.

Gator Learn

Subject

Login

[Home](#) [About](#) [Dashboard](#) [Create Tutor Listing](#) [Browse Listings](#) SFSU Software Engineering Project CSC 648-848, Fall 2025. For Demonstration Only

Gator Learn Sign Up

Full Name :

School Email:

@sfsu.edu

Password:

Comfirm Password:

(Optional) Pronouns:

(Optional) Photo:

Attach File

Accepts JPEG, PNG, WEBP

(Optional) Description:

☐

Check box to agree to [terms and conditions](#)

CANCEL

SIGN UP

Have an account already?

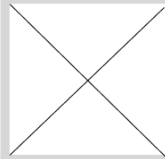
LOGIN

The site prompts Alex back to tutor listings.

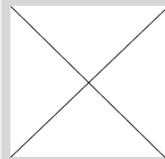
Current Listings

Found 10 tutors

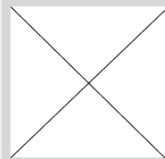
Sort By Price: \$ → \$\$\$\$ v

**Tutor Name**Tutoring for: Price: \$/hrTimes Available: [View more details>](#)

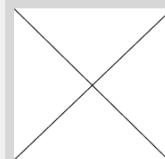
MESSAGE TUTOR

**Tutor Name**Tutoring for: Price: \$/hrTimes Available: [View more details>](#)

MESSAGE TUTOR

**Tutor Name**Tutoring for: Price: \$/hrTimes Available: [View more details>](#)

MESSAGE TUTOR

**Tutor Name**Tutoring for: Price: \$/hrTimes Available: [View more details>](#)

MESSAGE TUTOR

Use Case 2: Student logs in and wants to message a tutor

Sarah, an already registered user, logs back to the site to find another tutor for a different class



The screenshot shows the Gator Learn website interface. At the top, there is a dark header bar with the 'Gator Learn' logo on the left, a search bar in the center containing 'Subject ▼ eg. CSC 648', and a 'Login' button on the right. Below the header is a navigation bar with links: 'Home', 'About', 'Dashboard', 'Create Tutor Listing', and 'Browse Listings'. On the right side of the navigation bar, there is a small text string: 'SFSU Software Engineering Project CSC 648-848, Fall 2025. For Demonstration Only'. The main content area has a light gray background. In the center, the title 'Gator Learn Login' is displayed in a large, bold, black font, underlined. Below the title are two input fields: 'School Email:' followed by a text box containing '@sfsu.edu', and 'Password:' followed by an empty text box. Below these fields are two buttons: 'CANCEL' and 'LOGIN'. Further down, there is a link 'Forgot your password?' followed by a button labeled 'Click here'. At the bottom, there is a link 'Don't have an account?' followed by a button labeled 'SIGN UP'.

Sarah browses tutor listings, doesn't see something specific towards her, so she searches by her class and sorts by lowest price.

**Gator Learn**

Subject ▼ eg. CSC 648

Login

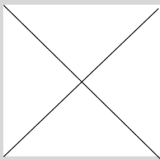
[Home](#) [About](#) [Dashboard](#) [Create Tutor Listing](#) [Browse Listings](#)

SFSU Software Engineering Project CSC 648-848, Fall 2025. For Demonstration Only

Search Results

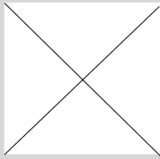
Results found: (#)

Sort By Price: \$ → \$\$\$\$ ▼



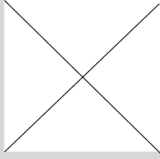
Tutor Name
Tutoring for:
Price: \$/hr
Times Available:

[View more details>](#)
[MESSAGE TUTOR](#)



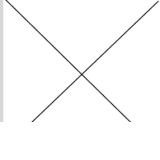
Tutor Name
Tutoring for:
Price: \$/hr
Times Available:

[View more details>](#)
[MESSAGE TUTOR](#)



Tutor Name
Tutoring for:
Price: \$/hr
Times Available:

[View more details>](#)
[MESSAGE TUTOR](#)



Tutor Name
Tutoring for:
Price: \$/hr
Times Available:

[View more details>](#)
[MESSAGE TUTOR](#)

Sarah looks at the tutor listing page to see a bit more detail on the tutor.



Gator Learn

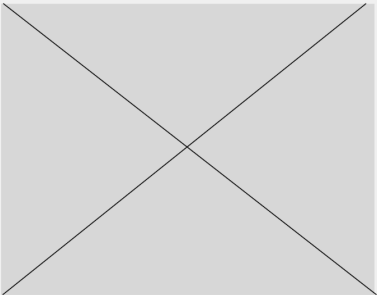
Subject ▼

Login



Home About Dashboard Create Tutor Listing Browse Listings SFSU Software Engineering Project CSC 648-848, Fall 2025. For Demonstration Only

Tutor Listing



Tutor Name is tutoring for class name

Rate: \$/hr

Availability:

List of availability

Description:

View Resume

View Sample Video

Send Tutor a Message

She thinks this tutor can help her so she presses the message tutor button. A form pops up to send a message to the tutor. She fills it out and sends it.


Gator Learn

Subject ▼

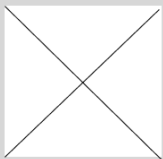
Log Out

[Home](#)
[About](#)
[Dashboard](#)
[Create Tutor Listing](#)
[Browse Listings](#)

SFSU Software Engineering Project CSC 648-848, Fall 2025. For Demonstration Only

Current Listings

Sort By Price: \$ → \$\$\$\$ v



Message : TUTOR NAME for CLASS NAME/NUM

Preferred Contact method:

Discord : Username123 or SFSU EMAIL

Comments:

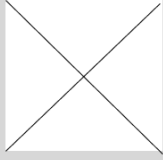
I have a midterm next tuesday. Can we meet online?

Cancel
Send Message

X

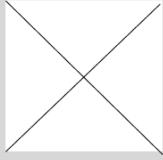
[details>](#)

TUTOR



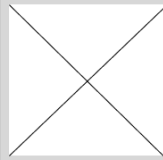
[details>](#)

TUTOR



[details>](#)

TUTOR



Tutor Name

Tutoring for:

Price: \$/hr

Times Available:

[View more details>](#)

MESSAGE TUTOR

Sarah can then check her sent messages on her dashboard. Tutor contacts her off-site.

Gator Learn

Search By ▼ Subject

Login

Home About Dashboard Create Tutor Listing Browse Listings

SFSU Software Engineering Project CSC 648-848, Fall 2025. For Demonstration Only

Messages

Sarah
Course Name/Number

XXX
Course Name/Number

XXX
Course Name/Number

XXX
Course Name/Number

Sarah
Course Name/Number

XXX
Course Name/Number

XXX
Course Name/Number

XXX
Course Name/Number

XXX
Course Name/Number

XXX

From: Sarah
Sub: Course Number/Number
Contact Information: (123) 456-7890

Comments:

I am having trouble on my milestone 2, do you think you can explain some concepts needed?

Sarah opens her message to the tutor.



SUBJECT LINE: Course Name/Number

To: James

From: Sarah (me)

Contact Information:

Phone number: (123) 456-7890

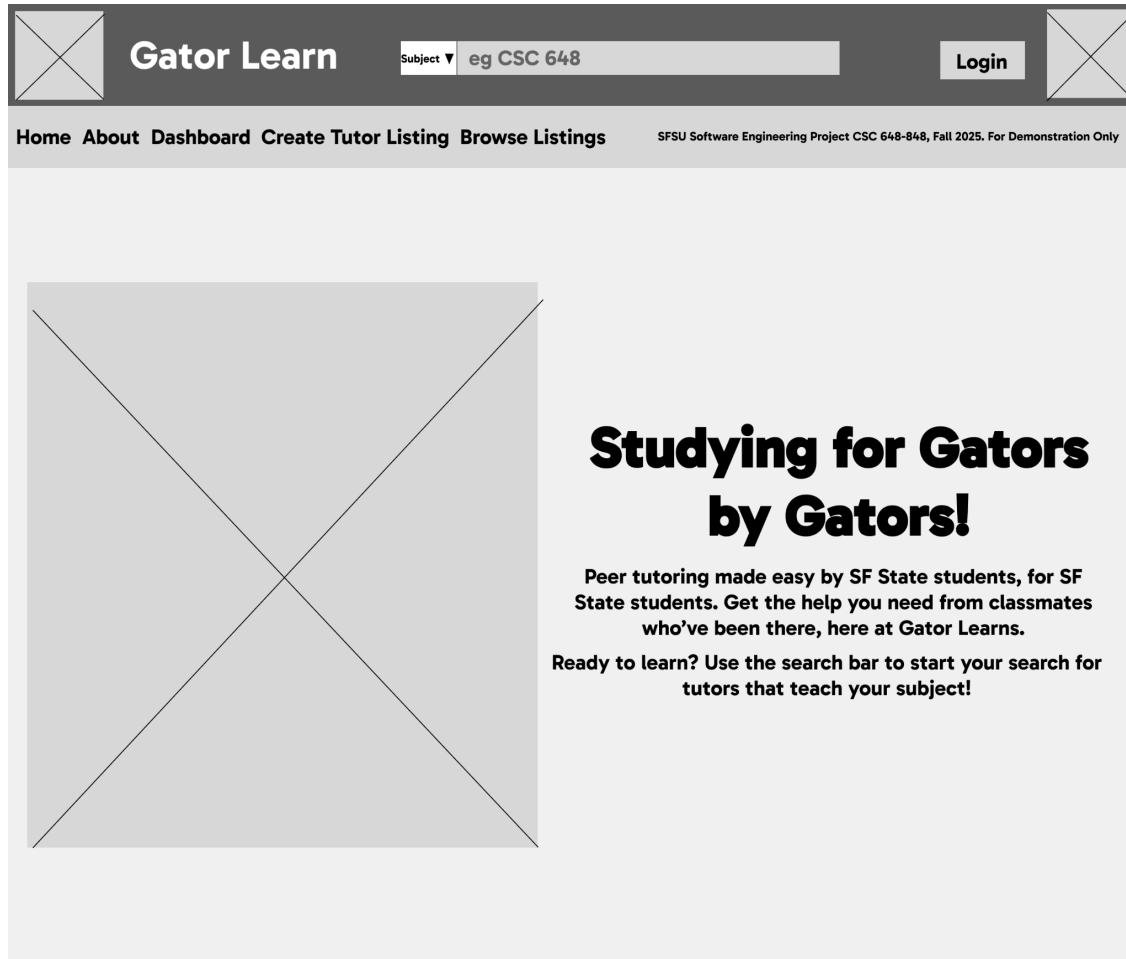
Comments:

**I am having trouble on my milestone 2,
do you think you can explain some
concepts needed?**

Close

Use Case 3: Student Fills Out Tutor Listing Form

James signs up and browses around.



He sees that there are few tutors for a course he can teach, so he fills out a tutor listing form.

Subject

Log out

Home About Dashboard Create Tutor Listing Browse Listings

SFSU Software Engineering Project CSC 648-848, Fall 2025. For Demonstration Only

Create Tutor Listing

For:

Class:

Price per hour: /hr

Available Times:

☒ Monday
☐ Tuesday
☐ Wednesday
☐ Thursday
☐ Friday
☐ Saturday
☐ Sunday

From:

AM
PM

To:

Description:

Optional:

Attach Resume/CV:
Accepts JPEG, PDF, WEBP

Attach Profile Image:
Accepts JPEG, PNG, WEBP

Attach Tutoring Video Sample:
Accepts MOV, MP4

CANCEL

SUBMIT

After James fills out his tutor listing, the site prompts him to wait a day or two for approval. He waits for approval.

Subject▼ eg. CSC 648

Login

[Home](#) [About](#) [Dashboard](#) [Create Tutor Listing](#) [Browse Listings](#) SFSU Software Engineering Project CSC 648-848, Fall 2025. For Demonstration Only

Create Tutor Listing

For: [SUBJECT given by search]

Class: CSC 648

Price per hour: \$/hr

Thank you for submitting your listing

Please wait 24 to 48 hours for approval message in your dashboard inbox. In the mean time, feel free to browse around if you need help on any courses.

Close

Optional:

Attach Resume/CV: Accepts JPEG, PDF, WEBP

Attach Profile Image: Accepts JPEG, PNG, WEBP

Attach Tutoring Video Sample: Accepts MOV, MP4

CANCEL

SUBMIT

He logs out in the meantime.

Gator Learn

Subject ▼ eg CSC 648

Login

Home About Dashboard Create Tutor Listing Browse Listings

SFSU Software Engineering Project CSC 648-848, Fall 2025. For Demonstration Only

Search Results

Results found: (#)

Sort By Price: \$ → \$\$\$\$ v

Tutor Name

Tutoring for:

View more details>

You are now logging out of Gator Learns.

Are you sure you want to log out?

YesNo

Tutor Name

Tutoring for:

Price: \$/hr

Times Available:

View more details>

Tutor Name

Tutoring for:

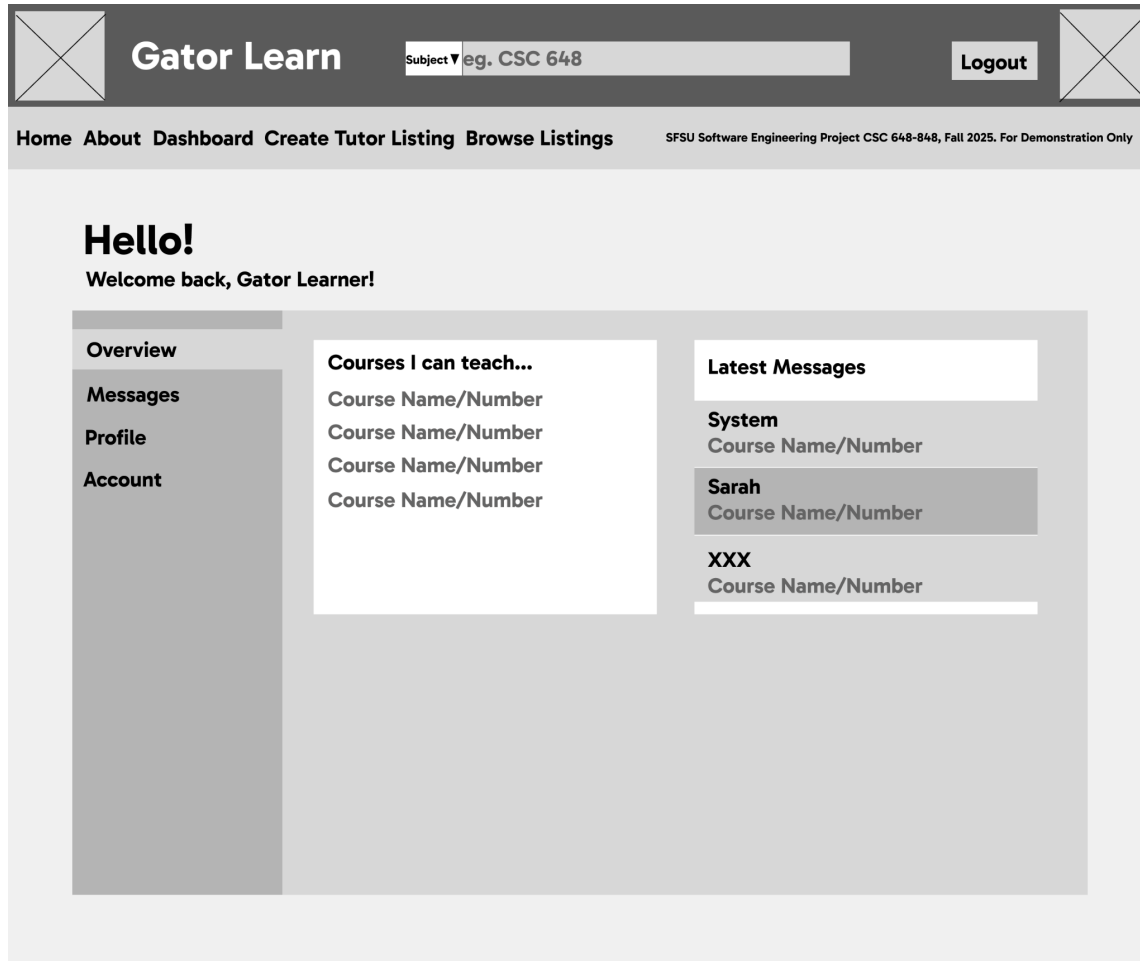
Price: \$/hr

Times Available:

View more details>

Use Case 4: The Student can see the dashboard for replies on the tutor listing and send messages

James logs back in and goes to his dashboard after waiting 2 days.



James sees he has a message from the system saying he was approved.



Gator Learn

Subject ▼ eg. CSC 648

Log Out



[Home](#) [About](#) [Dashboard](#) [Create Tutor Listing](#) [Browse Listings](#) SFSU Software Engineering Project CSC 648-848, Fall 2025. For Demonstration Only

SUBJECT LINE: Course Name/Number

To: Tutor/Student

From: System

Contact Information:

N/A

Comments:

**Congrats Student! Your listing was approved.
You will now receive messages in your inbox
from students**

Close

He also sees a few messages from students.

Gator Learn

Search By ▼ Subject

Logout

Home About Dashboard Create Tutor Listing Browse Listings

SFSU Software Engineering Project CSC 648-848, Fall 2025. For Demonstration Only

Messages

Sarah
Course Name/Number

XXX
Course Name/Number

XXX
Course Name/Number

XXX
Course Name/Number

Sarah
Course Name/Number

XXX
Course Name/Number

XXX
Course Name/Number

XXX
Course Name/Number

XXX
Course Name/Number

XXX

From: Sarah
Sub: Course Number/Number
Contact Information: (123) 456-7890

Comments:

I am having trouble on my milestone 2, do you think you can explain some concepts needed?

He clicks on one of the messages from the students and he chooses to read one and contact them off site.



Gator Learn

Subject ▼

Log Out



Home About Dashboard Create Tutor Listing Browse Listings

SFSU Software Engineering Project CSC 648-848, Fall 2025. For Demonstration Only

SUBJECT LINE: Course Name/Number

To:

From:

Contact Information:

Comments:

Use Case 5: Admin will use Workbench for their responsibilities.

5 Architecture and Database Organization

Database Organization:

1. UserAccount

- a. userID (PK)
- b. name
- c. email
- d. password
- e. photo
- f. description
- g. pronouns

2. TutorListing

- a. listingID (PK)
- b. userID (FK)
- c. subject (FK)
- d. course (FK)
- e. price
- f. availableTime
- g. resume
- h. description
- i. tutoringVideoSample
- j. live

3. Messages

- a. messageID (PK)
- b. listingID (FK)
- c. userID
- d. message
- e. phoneNumber
- f. dateAndTime

4. Subject (Will be used for pull-down, next to the search bar)

- a. subjectID (PK)
- b. subjectName
- c. category

- d. description

5. Courses

- a. courseID (PK)
- b. subjectID (FK)
- c. courseName
- d. courseNumber
- e. description

6. Feedback

- a. feedbackID (PK)
- b. listingID (FK)
- c. userID (FK)
- d. rating
- e. comment
- f. date

Media Storage:

We chose to store all uploaded media (images, CVs, and videos) as files on the server and only keep their file paths or URLs in the database. To ensure the security of these file references, we will implement input validation and sanitization.

This approach reduces database size, improves performance, and makes it easier to serve or update media content through a file system

Searching or Filtering:

We shall conduct searches and filters of tutor listings using MySQL queries.

For text comparison, the simple MySQL `LIKE` operator shall often be enough, however for more complex queries, we shall use the MySQL `REGEXP` operator.

For numerical comparison we shall use relational operators.

Some examples:

```
SELECT listingID FROM tutorListings WHERE course LIKE 'CSC%'
```

Selects all listingIDs whose corresponding listing is for a course string containing 'CSC'

```
SELECT * FROM tutorListings WHERE subject REGEXP '^.*Arts?.*$'
```

Selects all TutorListings containing 'Art' or 'Arts' in the subject

```
SELECT * FROM tutorListings WHERE price <= 20
```

Selects all TutorListings whose price is less than 20 dollars per hour

In addition, all database queries shall be done using parameterized queries to sanitize user input, avoiding possible SQL injection attacks.

Parameterized queries in Java can be implemented using [PreparedStatement](#)(s), for example:

```
// DB recognizes the ? in this query as "Fill in a single value here", so  
additional arguments to the query cannot be appended  
String queryStr = "SELECT * FROM tutorListings WHERE course LIKE ?";  
  
// Create the statement  
PreparedStatement statement = sqlConnection.prepareStatement(queryStr);  
  
// Set the first wildcard '?' in the query to be userInputCourseStr  
statement.setString(1, userInputCourseStr + "%");  
  
// Execute the query  
ResultSet result = preparedStatement.executeQuery();
```

6 Key Risks

Skills Risks

Git Proficiency

Risk: Some team members are still getting comfortable with branching, merging, and avoiding conflicts, which can slow collaboration or cause lost work.

Resolution: Conduct a short Git workflow session, enforce pull-request reviews into develop and main, and share resources and tutorial videos for participants to watch.

Database Management

Risk: Limited experience with MySQL schema design and remote database deployment could cause performance or data integrity problems.

Resolution: Assign a dedicated database lead, research and watch tutorials, and validate schema through local testing before deployment.

Schedule Risks

Meeting Commitments and Deadlines

Risk: Balancing the project workload with other coursework may lead to rushed deliverables near milestones.

Resolution: Use GitHub Projects for weekly task tracking, break milestones into smaller sections, and have check-ins to adjust workload distribution.

Technical Risks

Integrating All SFSU Courses and Majors

Risk: Populating an accurate list of all courses/majors may be time-consuming or technically challenging if scraped or manually entered.

Resolution: Start with a simplified subset (e.g., top departments), automate data retrieval with scripts if possible, and expand once basic functionality is stable.

Legal / Content Risks

Licensing and Third-Party Assets

Risk: Using UI libraries, icons, or course data without proper licensing could create copyright issues.

Resolution: Use only open-source MIT/BSD-licensed components and cite data sources (e.g., SFSU catalog).

7 Project Management

Our project is managed with the goal of meeting all deadlines, evenly distributing workload, and maintaining clear accountability across team members. We use **GitHub Projects** as our primary project management tool because it integrates directly with GitHub Issues and Pull Requests, making it simple to assign tasks, set deadlines, and track progress in one place. Each task is assigned to a responsible team member with a specific due date, and we conduct mid-point check-ins to ensure adherence to timelines. We also document our process in the repository so that all team members understand.

Team communication occurs through **Discord**, where we have dedicated channels for frontend, backend, git, and shared resources. We hold weekly one-and-a-half-hour calls (in addition to post-class meetings) to review progress, redistribute tasks as needed, and confirm our direction. Individual follow-ups are done when specific support or clarification is required.

For **Milestone 2**, we are using **Figma** to create wireframes and storyboards, enabling rapid prototyping and smooth transition to implementation. **Google Docs** is used for milestone documentation, providing real-time collaboration and version control that better suits group editing than binary Word files in Git. Following our instructor's guidance, the **frontend and backend teams work in parallel** while coordinating on shared API interfaces to improve efficiency and reduce dependencies.

8 GenAI Usage

Identifying key issues in project (Milo, ChatGPT): **MEDIUM**

Wrote out several key issues on my own and then had ChatGPT elaborate and find others. I found that it lacked proper solutions for our project and missed context that was necessary for the solutions.

Formatting milestone and documentation (Milo, ChatGPT): **HIGH**

After completing sections I asked ChatGPT to reformat it without changing the text. Was usually very helpful and I liked the formatting and stylistic choices that it made.

9 Team Lead Checklist

1. *So far all team members are fully engaged and attending team sessions when required.*

ISSUE: There have been a couple issues with completing deliverables on time or meetings missed, but we were able to resolve them internally.

2. *Team ready and able to use the chosen back and front end frameworks and those who need to learn are working on learning and practicing.*

ON TRACK

3. *Team reviewed suggested resources before drafting Milestone 2.*

ON TRACK

4. *Team lead checked Milestone 2 document for quality, completeness, formatting and compliance with instructions before the submission.*

DONE

5. *Team lead ensured that all team members read the final Milestone 2 document, and agree/understand it before submission.*

ON TRACK

6. *Team shared and discussed experience with genAI tools among themselves.*

DONE